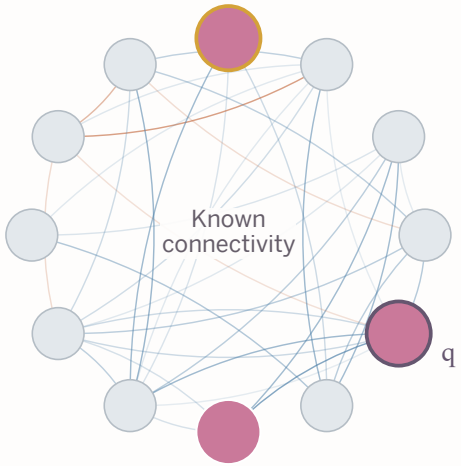
Predictive risk and recording decisions

in partially observed neural networks



100-unit networks · 5 recordings

Known connectivity; uncertain gains

- Recorded node
- Stimulated node
- Response readout (one of three)

1 Endpoint recording

Five recordings per allocation

Held-out response error

−98% 37 of 40 networks improved

Both ends vs neither, mean MSE

−98.13% [−99.19, −93.94]

Compatible response range

−99% 40 of 40 networks narrower

One dot per network

2 Error factors

Endpoint vs ordinary recording
Exact three-factor split

Input factor

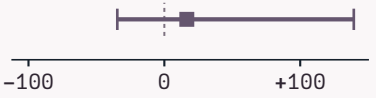
−98.7% [−99.5, −96.3]

Readout factor

−98.8% [−99.6, −96.9]

Recurrent factor

+16.5% [−34.6, +139.4]



Change in mean square (%)

95% paired network-bootstrap CI

3 Fifth recording

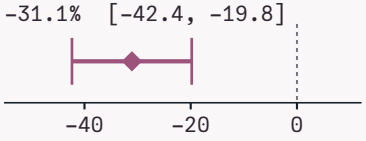
Endpoints kept; one site added
80 new independent networks

63 of 80 networks: better internal-combination estimates

Fifth site chosen for recurrent sums, versus a random fifth site; response-prediction gain uncertain

Combination estimation MSE

−31.1% [−42.4, −19.8]



Change vs random fifth site (%)

95% paired network-bootstrap CI

Integrated stimulus response

$$q = \int \Delta r(t)\, dt$$

$$\Delta r(t) = r_{\text{stim}}(t) - r_{\text{control}}(t)$$

Record endpoints for prediction; target the remaining internal information.

Every allocation uses **five recordings** and a common MAP fit; the stimulus response is withheld from calibration. Endpoint recording narrows compatible responses and contracts the input and readout error factors.

A fifth site chosen for recurrent gain sums improves their estimates on independent networks. Its response-prediction gain remains uncertain. All networks are simulated; no biological validation.